

AA-LOVASTATIN

Lovastatin Tablets USP 20mg

Lipid Metabolism Regulator

Pharmacology: Lovastatin is a cholesterol-lowering agent isolated from a strain of *Coniothyrium fuckelli*. After oral ingestion, lovastatin, which is an inactive lactone, is hydrolyzed to the corresponding β -hydroxyacid form. This principal metabolite is a specific inhibitor of 3-hydroxy-3-methyl-glutaryl-coenzyme A (HMG-CoA) reductase. This enzyme catalyzes the conversion of HMG-CoA to mevalonate, which is an early and rate-limiting step in the biosynthesis of cholesterol. Lovastatin reduces cholesterol production by the liver and induces some changes in cholesterol transport and disposition in the blood and tissues. The mechanism(s) of this effect is believed to involve both reduction of the synthesis of Low Density Lipoprotein (LDL), and an increase in LDL catabolism as a result of induction of the hepatic LDL receptors.

Human Pharmacology: Lovastatin has been shown to reduce both normal and elevated LDL-cholesterol concentrations. **The effect of lovastatin-induced changes in lipoprotein levels, including reduction of serum cholesterol, on cardiovascular morbidity or mortality, as well as on total mortality, has not been established.**

LDL is formed from VLDL and is catabolized predominantly by the high affinity LDL receptor. The mechanism of the LDL-lowering effect of lovastatin may involve both reduction of VLDL-cholesterol concentration, and induction of the LDL receptor leading to reduced production and/or increased catabolism of LDL-cholesterol.

Apolipoprotein B also falls substantially during treatment with lovastatin. Since each LDL particle contains one molecule of apolipoprotein B, and since little apolipoprotein B is found in other lipoproteins, this strongly suggests that lovastatin does not merely cause cholesterol to be lost from LDL, but also reduces the concentration of circulating LDL particles. However, a change in the composition of the LDL particle (lipid/protein ratio) cannot be excluded during treatment with lovastatin. In addition, lovastatin slightly increases HDL-cholesterol and reduces VLDL-cholesterol and plasma triglycerides.

The active form of lovastatin is a specific reversible inhibitor of HMG-CoA reductase, the enzyme which catalyzes the conversion of HMG-CoA to mevalonate. However, at therapeutic doses, the enzyme is not completely blocked, thereby allowing biologically necessary amounts of mevalonate to be available. Because the conversion of HMG-CoA to mevalonate is an early step in the biosynthetic pathway for cholesterol, therapy with lovastatin would not be expected to cause an accumulation of potentially toxic sterols.

Although cholesterol is the precursor of all steroid hormones, lovastatin, at therapeutic doses, has been shown to have no effect on steroidogenesis (see PRECAUTIONS, Endocrine Function).

Pharmacokinetics: Lovastatin is a lactone which is readily hydrolyzed in vivo to the corresponding β -hydroxyacid, a potent inhibitor of HMG-CoA reductase. Inhibition of HMG-CoA reductase is the basis for an assay in pharmacokinetic studies of the β -hydroxyacid metabolites (active inhibitors) and, following base hydrolysis, active plus latent inhibitors (total inhibitors) in plasma following administration of lovastatin.

Following an oral dose of ^{14}C -labelled lovastatin to man, 10% of the dose was excreted in urine and 83% in feces. The latter represents absorbed drug equivalents excreted in bile, as well as any unabsorbed drug. As a consequence of extensive hepatic extraction of lovastatin, the availability of drug to the general circulation is low and variable. In a single dose study in four hypercholesterolemic patients, it was estimated that less than 5% of an oral dose of lovastatin reaches the general circulation as active inhibitors. Following administration of lovastatin tablets, the coefficient of variation, based on between-subject variability, was approximately 40% for the area under the curve of total inhibitory activity in the general circulation.

Both lovastatin and its β -hydroxyacid metabolite are highly bound (>95%) to human plasma proteins. Animal studies demonstrated that lovastatin crosses the blood-brain and placental barriers.

The major active metabolites present in human plasma are the β -hydroxyacid of lovastatin, its 6'-hydroxy, 6'-hydroxymethyl, and 6'-exomethylene derivatives. Peak plasma concentrations of both active and total inhibitors were attained within 2 to 4 hours of dose administration. While the recommended therapeutic dose range is 20 to 80mg/day, linearity of inhibitory activity in the general circulation was established by a single dose study employing lovastatin tablet dosages from 60 to as high as 120mg.

With a once-a-day dosing regimen, plasma concentrations of total inhibitors over a dosing interval achieved a steady state between the second and third days of therapy and were about 1.5 times those following a single dose. When lovastatin was given under fasting conditions, plasma concentrations of both active and total inhibitors were on average about two-thirds those found when lovastatin was administered immediately after a standard test meal.

In a study of patients with severe renal insufficiency (creatinine clearance 0.167 - 0.5ml/s [10 - 30 ml/min]), the plasma concentrations of total inhibitors after a single dose of lovastatin were approximately two-fold higher than those in healthy volunteers.

Indications and Clinical Use: AA-LOVASTATIN (lovastatin) is indicated as an adjunct to diet, at least equivalent to the American Heart Association (AHA) Step 1 diet, for the reduction of elevated total and Low Density Lipoprotein Cholesterol (LDL-C) levels in patients with primary hypercholesterolemia (Types IIa and IIb), when the response to diet and other nonpharmacological measures alone has been inadequate.

After establishing that the elevation in plasma lipids represents a primary disorder not due to secondary conditions such as poorly-controlled diabetes mellitus, hypothyroidism, the nephrotic syndrome, liver disease, or dysproteinemias, it should be determined that patients for whom treatment with lovastatin is being considered have an elevated LDL-C level as the cause for an elevated total serum cholesterol. This may be particularly relevant for patients with total triglycerides over 4.52 mmol/L (400mg/dL) or with markedly elevated High Density Lipoprotein Cholesterol (HDL-C) values, where non-LDL lipoprotein fractions may contribute significantly to total cholesterol levels without apparent increase in cardiovascular risk. In general, LDL-C may be estimated according to the following equations:

$$\text{LDL-C (mmol/L)} = \text{Total cholesterol} - [(0.37 \times \text{triglycerides}) + \text{HDL-C}]$$

$$\text{LDL-C (mg/dL)} = \text{Total cholesterol} - [(0.16 \times \text{triglycerides}) + \text{HDL-C}]$$

When total triglycerides are greater than 4.52 mmol/L (400mg/dL) this equation is not applicable. In such patients, LDL-C may be obtained by ultra-centrifugation.

Lovastatin was also found to slow the progression of coronary atherosclerosis in patients with coronary heart disease as part of a treatment strategy to lower total and LDL-cholesterol to target levels. In two trials including this type of patient, i.e. in a secondary prevention intervention, lovastatin monotherapy was shown to slow the progression of coronary atherosclerosis as evaluated by computerized quantitative coronary angiography (QCA). This effect, however, was not accompanied by an improvement in the clinical endpoints (death, fatal/nonfatal myocardial infarction, hospitalization for unstable angina, and coronary revascularization procedure [PTCA and CABG] within the 2 - 2½ years' period of treatment. These trials, however, were not designed to demonstrate a reduction in the risk of coronary morbidity and mortality.

The effect of lovastatin on the progression of atherosclerosis in the coronary arteries has been corroborated by similar findings in another vasculature. In a trial including hyperlipidemic patients with early, asymptomatic carotid lesions and without known coronary artery disease, the effect of therapy with lovastatin on carotid atherosclerosis was assessed by B-mode ultrasonography. There was a significant regression of carotid lesions in patients receiving lovastatin alone compared to those receiving placebo alone. The predictive value of changes in the carotid vasculature for stroke has not yet been established. In the lovastatin group, there was a significant reduction in the number of patients with major cardiovascular events relative to the placebo group (5 vs 14) and a significant reduction in all-cause mortality (1 vs 8). This trial should be viewed as supportive and complementary to the others mentioned above. However, it was not powered to demonstrate a reduction in the risk of coronary morbidity and mortality. A larger trial of longer duration is needed to clarify the effect of lovastatin in monotherapy on clinical events.

Adverse Reactions:

Adverse reactions are listed as below:

Body as a whole: chest pain.

Gastrointestinal: acid regurgitation, dry mouth, vomiting.

Musculoskeletal: leg pain, shoulder pain, arthralgia.

Musculoskeletal disorders: Frequency not known: Immune-mediated necrotizing myopathy

Nervous system / psychiatric: insomnia, paresthesia.

Skin: alopecia, pruritus.
Special senses: eye irritation.

Nervous system disorders

Frequency not known:
myasthenia gravis

Eye disorders

Frequency not known:
ocular myasthenia

Effect on the Lens: (see PRECAUTIONS).

Post Marketing Experience: The following additional side effects have been reported since lovastatin was marketed: hepatitis, cholestatic jaundice, vomiting, anorexia, paresthesia neuropathy and psychic disturbances including anxiety, erythema multiform, including Stevens-Johnson syndrome; toxic epidermal necrolysis.

An apparent hypersensitivity syndrome has been reported rarely which has included one or more of the following features: anaphylaxis, angioedema, lupus-like syndrome, polymyalgia rheumatica, vasculitis, thrombocytopenia, leukopenia, eosinophilia, hemolytic anemia, positive ANA, ESR increase, arthritis, arthralgia, urticaria, asthenia, photosensitivity, fever, flushing, chills, dyspnea and malaise.

Undesirable effects: There have been rare post-marketing reports of cognitive impairment (e.g. memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use. These cognitive issues have been reported for all statins. The reports are generally non-serious and reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median 3 weeks). Increases in HbA1c and fasting blood glucose have been reported with statins. The risk of hyperglycemia, however, is outweighed by the reduction in vascular risk with statins.

Precautions: General: Before instituting therapy with AA-LOVASTATIN (lovastatin), an attempt should be made to control hypercholesterolemia with appropriate diet, exercise, weight reduction in overweight and obese patients, and to treat other underlying medical problems (see INDICATIONS AND CLINICAL USE). The patient should be advised to inform subsequent physicians of the prior use of lovastatin or any other lipid metabolism regulator.

Use in Homozygous Familial Hypercholesterolemia (FH): AA-LOVASTATIN is not effective or is less effective in patients with rare homozygous familial hypercholesterolemia, because these patients have no or very low levels of LDL receptor activity. Lovastatin appears to be more likely to raise serum transaminases in these homozygous patients.

Effect on the Lens: Current long-term data from clinical trials do not indicate an adverse effect of lovastatin on the human lens.

Effect on Lipoprotein(a) [Lp(a)]: In some patients, the beneficial effect of lowered total cholesterol and LDL cholesterol levels may be partly blunted by a concomitant increase in the Lp(a) levels. Until further experience is obtained from controlled clinical trials, it is suggested, where feasible, that Lp(a) measurements be carried out in patients placed on therapy with AA-LOVASTATIN.

Effect on CoQ10 Levels (Ubiquinone): A significant decrease in plasma CoQ10 levels in patients treated with lovastatin and other statins has been observed in short-term clinical trials. The clinical significance of a potential long-term statin-induced deficiency of CoQ10 has not yet been established.

Hypersensitivity: Although to date, hypersensitivity syndrome has not been described as such, in few instances eosinophilia and skin eruptions appear to be associated with lovastatin treatment. If hypersensitivity is suspected, AA-LOVASTATIN should be discontinued.

Use in Pregnancy: AA-LOVASTATIN is contraindicated during pregnancy.

Atherosclerosis is a chronic process and the discontinuation of lipid metabolism regulators during pregnancy should have little impact on the outcome of long-term therapy of primary hypercholesterolemia. Moreover, cholesterol and other products of the cholesterol biosynthesis pathway are essential components for fetal development, including synthesis of steroids and cell membranes. Because of the ability of inhibitors of HMG-CoA reductase such as lovastatin to decrease the synthesis of cholesterol and possibly other products of the cholesterol biosynthesis pathway, AA-LOVASTATIN may cause fetal harm when administered to a pregnant woman. A few reports have been received of congenital anomalies in infants whose mothers were treated during a critical period of pregnancy with HMG-CoA reductase inhibitors including lovastatin.

AA-LOVASTATIN should be administered to women of childbearing age only when such patients are highly unlikely to conceive. If the patient becomes pregnant while taking this drug, AA-LOVASTATIN

should be discontinued and the patient apprised of the potential hazard to the fetus.

Nursing Mothers: It is not known whether AA-LOVASTATIN is excreted in human milk. Because many drugs are excreted in human milk and because of the potential for serious adverse reactions in nursing infants from AA-LOVASTATIN, women taking AA-LOVASTATIN should not nurse their infants (see CONTRAINDICATIONS).

Pediatric Use: Limited experience is available in children. However, safety and effectiveness in children have not been established.

Geriatric Use: In patients over 60 years, efficacy appeared similar to that seen in the population as a whole, with no apparent increases in the frequency of clinical or laboratory adverse findings.

Use in Patients with Impaired Renal Function: Because lovastatin does not undergo significant renal excretion, modification of dosage should not be necessary in patients with moderate renal insufficiency. In patients with severe renal insufficiency (creatinine clearance < 0.5ml/s [30ml/min]), dosages above 20mg/day should be carefully considered and, if deemed necessary, implemented cautiously.

Endocrine Function: HMG-CoA reductase inhibitors interfere with cholesterol synthesis and as such might theoretically blunt adrenal and/or gonadal steroid production. Clinical studies with lovastatin have shown that this agent does not reduce plasma cortisol concentration or impair adrenal reserve, and does not reduce basal plasma testosterone concentration. However, the effects of HMG-CoA reductase inhibitors on male fertility have not been studied in an adequate number of patients. The effects, if any, on the pituitary-gonadal axis in premenopausal women are unknown.

Patients treated with lovastatin who develop clinical evidence of endocrine dysfunction should be evaluated appropriately. Caution should be exercised if an HMG-CoA reductase inhibitor or other agent used to lower cholesterol levels is administered to patients receiving other drugs (e.g. ketoconazole, spironolactone, or cimetidine) that may decrease the levels of endogenous steroid hormones.

Warnings: The effect of lovastatin-induced changes in lipoprotein levels, including reduction of serum cholesterol, on cardiovascular morbidity or mortality or total mortality has not been established.

Hepatic Effects: Marked persistent increases (to more than 3 times the upper limit of normal) in serum transaminases may occur in some adult patients who received lovastatin for at least one year (see ADVERSE REACTIONS under Laboratory Tests). When the drug was interrupted or discontinued in these patients, the transaminase levels fell slowly to pretreatment levels. The increases usually appeared 3 to 12 months after the start of therapy with lovastatin. In most cases, they were not associated with jaundice or other clinical signs or symptoms (see DRUG INTERACTIONS and Post Marketing Experience under ADVERSE REACTIONS).

It is recommended that liver function tests be performed at baseline and periodically thereafter in all patients. Particular attention should be paid to patients who develop elevated serum transaminase levels, and in these patients, measurements should be repeated promptly and then performed more frequently. If the transaminase levels show evidence of progression, particularly if they rise to three times the upper limit of normal and are persistent, the drug should be discontinued.

The drug should be used with caution in patients who consume substantial quantities of alcohol and/or have a past history of liver disease. Active liver disease or unexplained serum transaminase elevations are contraindications to the use of lovastatin; if such conditions develop during therapy, the drug should be discontinued.

Moderate elevations of serum transaminases (less than three times the upper limit of normal) have been reported following therapy with lovastatin (see ADVERSE REACTIONS). These changes were not specific to lovastatin and were also observed with comparative lipid metabolism regulators. They generally appeared with the first 3 months after initiation of therapy, were often transient and were not accompanied by any other symptoms. They did not necessitate interruption of treatment.

Muscle Effects: *CPK:* Transient elevations of creatine phosphokinase (CPK) levels are commonly seen in lovastatin-treated patients, but have usually been of no clinical significance.

Myalgia and muscle cramps have also been associated with lovastatin therapy.

Myopathy has occurred rarely and should be considered in any patient with diffuse myalgias, muscle tenderness or weakness and/or marked elevation of creatine phosphokinase (10 times the upper limit of normal). There have been reports of severe rhabdomyolysis that precipitated acute renal failure. Therapy with lovastatin should be discontinued if marked elevation of CPK levels occurs, or if myopathy is diagnosed or suspected.

Rhabdomyolysis with or without renal impairment has been reported in seriously ill patients receiving erythromycin concomitantly with lovastatin.

Some patients on immunosuppressive therapy, including cyclosporine, developed myopathy within a year after starting therapy with lovastatin. The incidence of myopathy has also been reported in patients receiving lovastatin concomitantly with gemfibrozil or niacin. Most of these patients were taking lovastatin 40 - 80mg/day.

Therefore, the benefits and risks of using AA-LOVASTATIN concomitantly with immunosuppressive drugs, erythromycin, fibrates or lipid-lowering doses of niacin should be carefully considered.

Because of an apparent relationship between increased plasma levels of active metabolites derived from lovastatin and myopathy, the daily dosage in patients taking immunosuppressants should not exceed 20mg/day (see DOSAGE AND ADMINISTRATION). Even at this dosage, the benefits and risks of using lovastatin in patients taking immunosuppressants should be carefully considered.

In patients receiving cyclosporine, lovastatin should be temporarily discontinued if systemic azole derivative antifungal therapy is required; patients not taking cyclosporine should be carefully monitored if systemic azole derivative antifungal therapy is required.

Interruption of therapy with AA-LOVASTATIN should be considered in any patient with an acute, serious condition, suggestive of a myopathy or having a risk factor predisposing to the development of renal failure or rhabdomyolysis, such as severe acute infection, hypotension, major surgery, trauma, severe metabolic, endocrine or electrolyte disorders and uncontrolled seizures.

Patients should be advised to report promptly unexplained muscle pain tenderness or weakness, particularly if accompanied by malaise or fever.

There have been very rare reports of an immune-mediated necrotizing myopathy (IMNM) during or after treatment with some statins. IMNM is clinically characterized by:

- persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment;
- muscle biopsy showing necrotizing myopathy without significant inflammation;
- improvement with immunosuppressive agents.

Colchicine: Cases of myopathy, including rhabdomyolysis, have been reported with lovastatin coadministered with colchicine, and caution should be exercised when prescribing lovastatin with colchicine.

Myasthenia Gravis/ Ocular Myasthenia

In few cases, statins have been reported to induce de novo or aggravate pre-existing myasthenia gravis/ocular myasthenia. AA-Lovastatin should be discontinued in case of aggravation of symptoms. Recurrences when the same or a different statin was (re-)administered have been reported.

Contraindications:

- Hypersensitivity to any component of this medication.
- Active liver disease or unexplained persistent elevations of serum transaminases.
- Pregnancy and lactation (see also PRECAUTIONS).
- Concomitant administration of strong CYP3A4 inhibitors (e.g. itraconazole, ketoconazole, posaconazole, voriconazole, HIV protease inhibitors, boceprevir, telaprevir, erythromycin, clarithromycin, telithromycin and nefazodone).
- Concomitant administration of cyclosporine.

Drug Interactions:

Concomitant Therapy with Other Lipid Metabolism Regulators: Combined drug therapy should be approached with caution as information from controlled studies is limited.

Bile Acid Sequestrants: Preliminary evidence suggests that the cholesterol-lowering effects of lovastatin and the bile acid sequestrant, cholestyramine, are additive.

When AA-LOVASTATIN is used concurrently with cholestyramine or any other resin, an interval of at least two hours should be maintained between the two drugs, since the absorption of lovastatin may be impaired by the resin.

Gemfibrozil, Fenofibrate and Niacin: Myopathy, including rhabdomyolysis, has occurred in patients who were receiving co administration of lovastatin with fibric acid derivatives and niacin, particularly in subjects with pre-existing renal insufficiency (see WARNINGS). Concurrent use of fibrates may cause severe myositis and myoglobinuria.

Erythromycin: see WARNINGS: Muscle Effects.

Angiotensin-Converting Enzyme Inhibitors: Hyperkalemia associated with myositis (myalgia and elevated CPK) has been reported in the case of a single patient with insulin-dependent diabetes mellitus and mild renal insufficiency who received lovastatin concomitantly with an angiotensin-converting enzyme inhibitor (lisinopril).

Coumarin Anticoagulants: Clinically evident bleeding and/or increased prothrombin time have been reported occasionally in patients taking coumarin anticoagulants concomitantly with lovastatin. Careful monitoring of prothrombin time is therefore recommended in such cases.

Digoxin: In patients with hypercholesterolemia, concomitant administration of lovastatin and digoxin resulted in no effect on digoxin plasma concentrations.

Beta-Adrenergic Blocking Drugs: In healthy volunteers, the co administration of propranolol and lovastatin resulted in a slight decrease of the AUC of lovastatin and its metabolites as well as in a significant decrease of the C_{max} for the lovastatin metabolites. The clinical interpretation of this phenomenon is difficult as it may indicate a greater uptake of lovastatin by the liver.

There was no clinically relevant interaction reported in patients who have been receiving lovastatin concomitantly with beta-adrenergic blocking agents.

Antipyrine: Antipyrine was used as a model for drugs metabolized by the microsomal hepatic enzyme system (cytochrome P450 system). Lovastatin had no effect on the pharmacokinetics of antipyrine.

Other Concomitant Therapy: Caution should be exercised with concomitant use of immunosuppressants and itraconazole (see WARNINGS).

Although specific interaction studies were not performed, in clinical studies, lovastatin was used concomitantly with calcium-channel blockers (such as verapamil HCl, nifedipine and diltiazem HCl), a number of diuretics and nonsteroidal anti-inflammatory drugs (NSAIDs), hypoglycemic drugs (chlorpropamide, glyburide, insulin), without evidence of clinically significant adverse interactions.

Drug/Laboratory Test Interactions: Lovastatin may elevate creatine phosphokinase and transaminase levels (see Laboratory Tests under ADVERSE REACTIONS). In the differential diagnosis of chest pain in a patient on therapy with lovastatin, cardiac and non-cardiac fractions of these enzymes should be determined.

Contraindicated Drugs

Strong inhibitors of CYP3A4: Concomitant use with strong CYP3A4 inhibitors (e.g. itraconazole, ketoconazole, posaconazole, voriconazole, HIV protease inhibitors, boceprevir, telaprevir, erythromycin, clarithromycin, telithromycin and nefazodone) is contraindicated.

Cyclosporine: The risk of myopathy/rhabdomyolysis is increased by concomitant administration of cyclosporine. Concomitant use of this drug with lovastatin is contraindicated.

Other Drugs

- Gemfibrozil, other fibrates, niacin $\geq 1\text{g/day}$:
These drugs increase the risk of myopathy when given concomitantly with lovastatin, probably because they can produce myopathy when given alone. There is no evidence to suggest that these agents affect the pharmacokinetics of lovastatin. Myopathy, including rhabdomyolysis, has occurred in patients who were receiving coadministration of lovastatin with fibric acid derivatives or niacin.
- Danazol, verapamil, diltiazem: The risk of myopathy/rhabdomyolysis is increased by concomitant administration of danazol, verapamil, diltiazem particularly with higher doses of lovastatin.
- Amiodarone: The risk of myopathy/rhabdomyolysis is increased when amiodarone is used concomitantly with higher doses of a closely related member of the HMG-CoA reductase inhibitor class.
- Colchicine: Cases of myopathy, including rhabdomyolysis, have been reported with lovastatin coadministered with colchicine, and caution should be exercised when prescribing lovastatin with colchicine.

Route of Administration: Oral

Dosage and administration: The patient should be placed on at least an equivalent of the American Heart Association (AHA) Step 1 diet before receiving AA-LOVASTATIN (lovastatin) and should continue on this diet during treatment with AA-LOVASTATIN. If appropriate, a program of weight control and physical exercise should be implemented.

Patients with Hypercholesterolemia: The usual starting dose is 20mg/day given as a single dose with the evening meal. Single daily doses given with the evening meal have been shown to be more effective than the same dose given with the morning meal, perhaps because cholesterol is synthesized mainly at night. Adjustments of dosage, if required, should be made at intervals of not less than 4 weeks, to a maximum of 80mg daily given in single doses or divided doses with the morning and evening meals. Divided doses (i.e. twice daily) tend to be slightly more effective than single daily doses.

Cholesterol levels should be monitored periodically and consideration should be given to reducing the dosage of AA-LOVASTATIN if cholesterol levels fall below the targeted range, such as that recommended by the Second Report of the U.S. National Cholesterol Education Program (NCEP).

Patients with Established Coronary Heart Disease: In the trials involving patients with coronary heart disease and administered lovastatin with (colestipol) or without concomitant therapy, the dosages used were 20 to 80mg daily, given in single or divided doses. In the two trials which utilized lovastatin alone, the dose was reduced if total plasma cholesterol decreased to below 2.85mmol/L or if LDL-cholesterol decreased to below 2.1mmol/L, respectively.

Concomitant Therapy: See Drug Interactions, Concomitant Therapy with Other Lipid Metabolism Regulators.

In patients taking immunosuppressive drugs concomitantly with lovastatin, the maximum recommended dosage of AA-LOVASTATIN is 20mg/day (see WARNINGS, Muscle Effects). Concomitant Therapy

The combined use of lovastatin with gemfibrozil should be avoided.

In patients taking danazol, verapamil, diltiazem, fibrates (except gemfibrozil) or lipid-lowering dose of niacin ($\geq 1\text{g/day}$) concomitantly with AA-Lovastatin, the dose of AA-Lovastatin should not exceed 20mg/day.

In patients taking amiodarone concomitantly with AA-Lovastatin, the dose of AA-Lovastatin should not exceed 40mg/day.

Symptoms and Treatment of Overdosage: Five healthy human volunteers have received up to 200mg of lovastatin as a single dose without clinically significant adverse experiences. A few cases of accidental overdosage have been reported; no patients had any specific symptoms and all patients recovered without sequelae. The maximum dosage taken was 5 - 6g.

In the event of overdosage, treatment should be symptomatic and supportive, liver function should be monitored, and appropriate therapy instituted. Until further experience is obtained, no specific therapy of overdosage can be recommended.

The dialyzability of lovastatin and its metabolites in man is not known.

Availability of Dosage Forms:

AA-LOVASTATIN 20mg tablets are light blue, octagonal, flat-faced, bevelled-edge tablets, engraved "APO" on one side, scored and engraved "LOVA" over "20" on the other side. Available in blister pack of 30 and 500 tablets.

Storage Conditions:

Store below 30°C. Protect from light.

Manufacturer: Apotex Inc, 150 Signet Drive, Weston (Toronto), Ontario, Canada, M9L 1T9.

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